Introduction
Retail businesses generate large volumes of transactional data daily. Analyzing this data helps identify profit trends, customer behavior, and inventory performance. This project focuses on evaluating retail business performance to detect profit-draining categories, improve inventory turnover, and understand seasonal product demand.

Abstract
The project analyzes retail transaction data using SQL, Python, and Tableau to uncover key business insights. SQL is used for data cleaning and profit margin calculations, while Python (Pandas) is used for correlation analysis between inventory turnover and profitability. Tableau is used to build an interactive dashboard for visualization. The results help identify underperforming products, seasonal trends, and opportunities for improving business efficiency and profitability.

Tools Used
SQL – Data cleaning and querying
Python (Pandas, Seaborn) – Data analysis and visualization
Tableau – Interactive dashboard creation

Steps Involved in Building the Project
1. Data Collection & Import
Imported retail transactional dataset into SQL database.
2. Data Cleaning (SQL)
Removed null values and duplicates
Standardized column formats (dates, categories, prices)
3. Profitability Analysis (SQL)
Calculated profit margin using:
Profit Margin = (Profit / Sales) * 100
Grouped data by category and sub-category
4. Correlation Analysis (Python)
Used Pandas to analyze relationship between inventory days and profitability
Identified whether slow-moving inventory affects profit
5. Data Visualization (Tableau)
Built dashboard with filters: Region, Product Category, Season
Used bar charts, line graphs, and heatmaps
6. Insight Generation
Identified low-profit categories
Detected seasonal demand patterns
Found overstocked and slow-moving items

Key Insights
Certain product categories generate high sales but low profit margins
Slow-moving inventory negatively impacts profitability
Seasonal trends significantly affect product demand
Some regions perform better due to higher sales efficiency

Conclusion
This project demonstrates how data analysis can improve retail business performance. By identifying profit-draining categories and inefficient inventory management, businesses can make informed decisions. The use of SQL, Python, and Tableau provides a complete pipeline from data processing to visualization. Future improvements can include predictive analysis for demand forecasting.